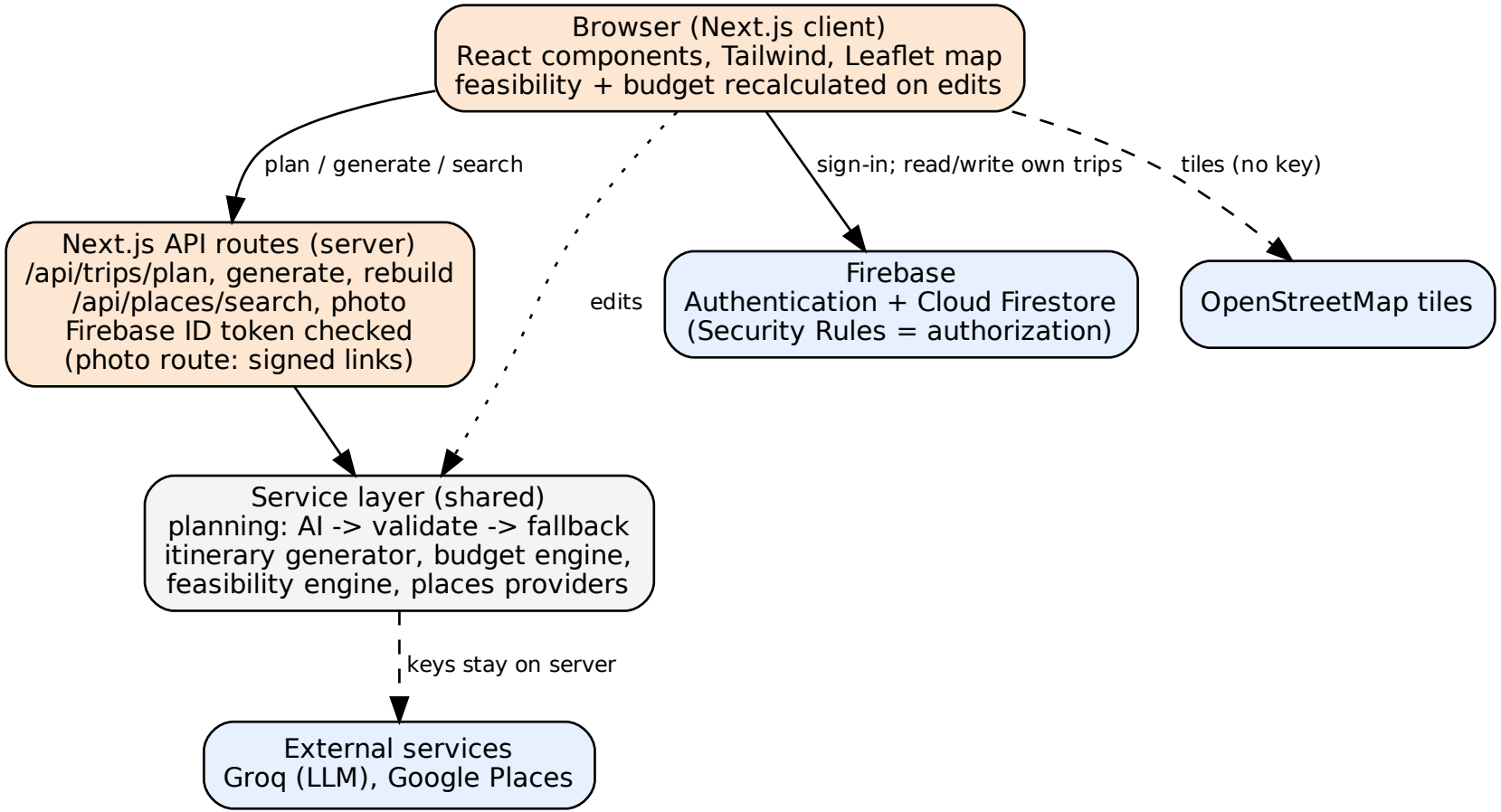
Browser (Next.js client)
React components, Tailwind, Leaflet map
feasibility + budget recalculated on edits
plan / generate / search
sign-in; read/write own trips
tiles (no key)
Next.js API routes (server)
/api/trips/plan, generate, rebuild
/api/places/search, photo
Firebase ID token checked
(photo route: signed links)
edits
Firebase
Authentication + Cloud Firestore
(Security Rules = authorization)
OpenStreetMap tiles
Service layer (shared)
planning: AI -> validate -> fallback
itinerary generator, budget engine,
feasibility engine, places providers
keys stay on server
External services
Groq (LLM), Google Places